

Other components

Wind tower. Wind tower is a traditional architectural element used to pre-cool the air and direct it into interiors of a building.

MERV filters. Minimum efficiency reporting value (MERV) is a number from 1 to 16 that is relative to an air filter's efficiency. The higher the MERV, the more efficient the air filter is at removing particles from the outside air.

Geothermal systems. A geothermal system is a non-conventional heating and/or cooling system that pumps heat to or from the ground crust. This system takes advantage of the moderate temperatures in the ground to boost efficiency and reduce the operational costs of heating and cooling systems.

High-performance glass. High-performance glass reduces the entry of heat while allowing the penetration of sunlight simultaneously. The glass comes in three types: insulated (double- or triple-glazed) glass, inert-gas-filled glazing and low-emissivity glass.

Wall and roof insulation. The heat balance of a building reveals that at least 15 to 20 per cent of the heat ingress into a building can be through walls and roof. Hence insulating walls and roof is extremely critical in the energy performance of a building.



Bamboo-based furniture. Bamboo is increasing in popularity as an alternative to conventional wood such as timber and teak. It is regarded as a rapidly renewable resource because of its faster harvesting cycle.



Intelligent building management

Energy, Delta Power and many other firms have installed technologies such as solar PV and building-integrated solar photovoltaics, which have become an integral part of the building envelope.

The green building movement in India has provided avenues to technology providers for developing out-of-the-box models that are ecologically superior and economically viable. Blue Star, Carrier, Daikin Industries, Emerson, Schneider, Trane India and Voltas are some of major energy-efficient heating, ventilation and air-conditioning (HVAC) system providers. The incremental costs of green building technologies have come down significantly.

Focus of R&D

Light pipes can be incorporated into architecture for transporting or distributing daylight or artificial light, minimising the loss of light. Daylight harvesting reduces the use of artificial light inside a building when natural daylight is available, helping cut down on energy consumption. Advanced online energy and water monitoring systems can be used to monitor the efficient use of various systems that consume, distribute or generate energy and water.

R&D is happening in these areas:

1. Computational fluid dynamics analysis, which aims to further enhance the efficiency of HVAC
2. Net-zero-energy, high-performance buildings—buildings that generate as much energy through renewable means as is consumed by the building
3. Water-positive buildings that regenerate more water than they consume
4. Life-cycle analysis to assess environmental impacts associated with all the stages of a product's life from start to end
5. Complete retrofitting of existing buildings by renovating energy-wasting homes and offices and bringing them up to passive house standards. ●